

Lecture 1: Introduction to Signal Processing

ADS203TC - Signal and Image Processing

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After this lecture, the student will be able to:

- **Define** discrete-time signals and systems using formal mathematical notation.
- **Classify** signals based on fundamental properties such as periodicity and energy.
- **Characterize** systems according to the principles of linearity, time-invariance, causality, and stability.
- **Identify and describe** applications of signal processing across various scientific and engineering disciplines.

Outline

- 1 Introduction
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- 7 Energy and Power Signals
- 8 Systems and Properties
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What is Signal Processing?

Signal Processing encompasses the theory and application of filtering, coding, transmitting, estimating, detecting, analyzing, recognizing, synthesizing, recording, and reproducing signals.

- **Analog Signal Processing (ASP):** Using passive/active electronic circuits (Resistors, Capacitors, Op-Amps).
- **Digital Signal Processing (DSP):** Using general-purpose computers or dedicated hardware to manipulate signal numbers.

Applications of DSP

Digital Signal Processing is ubiquitous across multiple scientific and engineering disciplines:

- **Communications:** Data compression, error correction, multiplexing, modulation (e.g., 5G/6G, Wi-Fi).
- **Control Systems:** Digital controllers, sensor data fusion, robotics, automotive engine control.
- **Biomedical Engineering:** ECG/EEG analysis, medical imaging (MRI, CT scans), hearing aids.
- **Data Science & Computing:** Time-series forecasting, audio/speech recognition (AI), financial market analysis.

Main concept

DSP provides tools to analyze, transform, and extract information from data (that can be considered as *signals*) before it is used in statistical or Machine Learning (ML) models.

1. Time-Series Analysis & Preprocessing

- **Decomposition:** Separate a signal into constituent parts like trend, seasonality (periodicity), and residual noise.
- **Filtering & Smoothing:** Remove noise and artifacts to reveal underlying patterns.
- **Spectral Analysis:** Employ the Fourier Transform to identify dominant frequencies, essential for understanding cyclical behavior (e.g., daily web traffic).

Why do we use Digital Signal Processing?

Digital Signal Processing (DSP) offers several advantages over traditional analog methods:

1 Programmability (Flexibility):

- A DSP system can be reconfigured by changing the software.
- Analog systems require hardware rewiring (redesigning the circuit board).

2 Accuracy and Precision:

- Analog components have tolerances ($R \pm 5\%$) and drift with temperature/age.
- Digital systems (0s and 1s) are perfectly reproducible and deterministic.

3 Storage and Transport:

- Digital signals can be stored on magnetic/optical media without degradation.
- Analog signals degrade with every copy (noise floor increases).

4 Sophistication:

- Can implement complex algorithms (e.g., adaptive filtering, encryption) that are physically impossible with analog hardware.

Limitations of DSP

Despite the advantages, DSP is not infinite:

- **Bandwidth / Speed:**

- Limited by the speed of A/D and D/A converters.
- Extremely high frequencies (Radar, Microwave) still require analog front-ends.

- **Finite Word Length Effects:**

- We cannot represent numbers with infinite precision.
- Round-off errors and quantization noise.

Brief description of a DSP System

Most practical DSP systems interact with the real (analog) world to analyze and process signals.



- **A/D:** Converts analog signal to digital sequence.
- **DSP:** Performs the mathematical operations.
- **D/A:** Converts processed sequence back to analog.
- $x_a(t)$ and $y_a(t)$ are the input and output analog signals.
- $x(n)$ and $y(n)$ are the input and output digital (or discrete) signals.

What is a Signal?

Definition 1.1.1 (Signal)

A signal is any physical quantity that varies with time, space, or any other independent variable or variables, formally modeled as a mathematical function.

Core Function:

- Signals carry information about the state or behavior of a physical system.
- Mathematically, we describe a signal as a function of one or more independent variables.

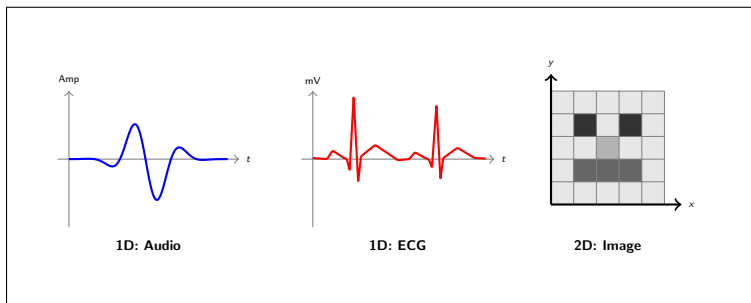
Examples of Signals

1D Signals (Time):

- Speech (Sound pressure vs. time)
- ECG (Electrical activity vs. time)

Multidimensional Signals:

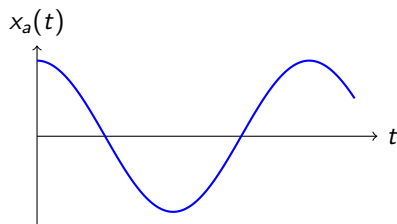
- Images (Brightness vs. x, y)
- Video (Intensity vs. x, y, t)



Time Domain and Amplitude

Continuous-Time (t)

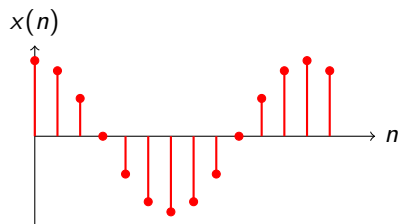
- Defined for all $t \in \mathbb{R}$.
- Notation: $x_a(t)$ or $x(t)$.



Continuous-Time

Discrete-Time (n)

- Defined only at specific instants $n \in \mathbb{Z}$.
- Notation: $x(n)$ or $x[n]$.



Discrete-Time

The "Digital" Signal*

Definition 1.1.2 (Digital Signal)

A strictly **Digital Signal** is discrete in both:

- 1 **Time** (Discrete-time)
- 2 **Amplitude** (Discrete-valued/Quantized)

Important Distinction

In theoretical DSP analysis, we often assume signals are **discrete-time** but **continuous-valued** to simplify the mathematics. We treat amplitude quantization errors as noise later in the design process.



Multichannel vs. Single-channel:

- **Single-channel:** One sensor (e.g., standard microphone).
- **Multichannel:** Vector of signals $\mathbf{S}(t) = [s_1(t), s_2(t), s_3(t)]^T$ (e.g., 3-lead ECG, seismograph).

Deterministic vs. Random:

- **Deterministic:** Exquisitely described by a mathematical expression (e.g., $x(t) = 5 \sin(20\pi t)$). Past, present, and future are known.
- **Random:** Cannot be precisely predicted. Described via statistics (mean, variance). (e.g., Thermal noise).

Exercise 1.1.1: Signal Classification

Problem: Classify the following signals according to whether they are:

- (a) Continuous-time or Discrete-time.
- (b) Continuous-valued or Discrete-valued (Digital).
- (c) Deterministic or Random.

Signals to classify:

- ① $x(n) = \sin(0.1\pi n)$ stored as a 16-bit float.
- ② The daily closing price of a specific stock over one year.
- ③ $x(t) = e^{-2t}u(t)$ (Voltage across a discharging capacitor).

Solution 1.1.1: Signal Classification

1. $x(n) = \sin(0.1\pi n)$ stored as a 16-bit float.

- **Discrete-time:** Defined on integer n .
 - **Discrete-valued:** "16-bit" implies quantization (finite precision).
 - **Deterministic:** Defined by a mathematical function.
-

2. Daily closing price of a stock.

- **Discrete-time:** Defined at specific intervals (days).
 - **Discrete-valued:** Money has a smallest unit (e.g., cents), though often treated as continuous for analysis.
 - **Random:** Cannot be predicted precisely by a formula.
-

3. $x(t) = e^{-2t}u(t)$.

- **Continuous-time:** Defined for all t .
- **Continuous-valued:** Can take any real value.
- **Deterministic:** Defined by a strict formula.



- **Signal & DSP:** Signals carry information; DSP manipulates them digitally for flexibility, precision, and storage.
- **Applications:** Integral to communications, control, biomedical, and data science.
- **DSP System:** Analog Input $\rightarrow$ ADC $\rightarrow$ Processor $\rightarrow$ DAC $\rightarrow$ Analog Output.
- **Classifications:** Continuous vs. Discrete (Time/Amplitude), Multichannel vs. Single Channel, Deterministic vs. Random.
- **Digital Definition:** Discrete in time AND discrete in amplitude.

Continuous-Time Sinusoidal Signals

The fundamental building block of **analog** signal analysis is the sinusoid:

$$x_a(t) = A \cos(\Omega t + \theta) = A \cos(2\pi F t + \theta)$$

Parameters:

- A : Amplitude (Real)
- Ω : Angular Frequency (rad/sec)
- F : Frequency (cycles/sec or Hz)
- θ : Phase (radians)
- Relationship: $\Omega = 2\pi F$

Complex Exponential Form:

$$x_a(t) = A e^{j(\Omega t + \theta)}$$

(Using Euler's Identity)

Key Properties:

- 1 **Property 1.2.1 (Periodicity)**: Periodic with $T_p = 2\pi/\Omega$ for any Ω .
- 2 **Property 1.2.2 (Uniqueness)**: Distinct frequencies $\Omega_1 \neq \Omega_2$ yield distinct signals.
- 3 **Property 1.2.3 (Oscillation Rate)**: Increasing Ω strictly increases oscillation rate.

Complex Exponential Form of Sinusoidal Signals

Complete Representation

$$\begin{aligned}x_a(t) &= A \cos(\Omega t + \theta) \\&= \Re\{Ae^{j(\Omega t + \theta)}\} \\&= \frac{A}{2} [e^{j(\Omega t + \theta)} + e^{-j(\Omega t + \theta)}]\end{aligned}$$

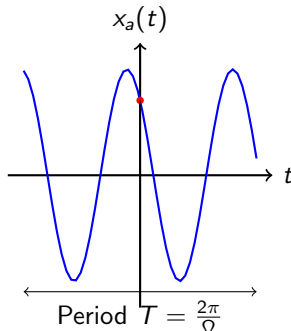
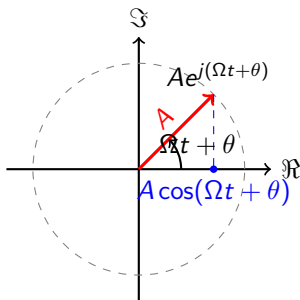
Geometric Interpretation: Rotating phasor in complex plane with:

- Magnitude A
- Instantaneous phase $\Omega t + \theta$
- Projection on real axis yields sinusoid

$$\underbrace{A \cos(\Omega t + \theta)}_{\text{Real Sinusoid}} \iff \underbrace{\frac{A}{2} e^{j\theta} e^{j\Omega t}}_{\text{Positive Frequency}} + \underbrace{\frac{A}{2} e^{-j\theta} e^{-j\Omega t}}_{\text{Negative Frequency}}$$

Geometric Interpretation of Complex Exponential Form

Phasor Representation: $A \cos(\Omega t + \theta) = \Re\{Ae^{j(\Omega t + \theta)}\}$



(a) Rotating phasor in the complex plane (b) Resulting real sinusoid

The rotating phasor's projection onto the real axis generates the real sinusoidal signal.

Key insight: A single complex exponential $e^{j(\Omega t + \theta)}$ rotates counterclockwise at frequency Ω , while its complex conjugate $e^{-j(\Omega t + \theta)}$ rotates clockwise. Their sum (scaled by $A/2$) produces the real sinusoid (see *Lecture 1 - demos.ipynb on LMC*).

Discrete-Time Sinusoidal Signals*

A discrete-time sinusoid is defined strictly at integer indices n :

$$x(n) = A \cos(\omega n + \theta) = A \cos(2\pi f n + \theta)$$

Notation Shift

- ω : Discrete angular frequency (radians/sample).
- f : Discrete frequency (cycles/sample).
- Relationship: $\omega = 2\pi f$.

Key Question: Do discrete sinusoids behave exactly like continuous ones?

Answer: No! There are three fundamental differences.

Property 1.2.4: DT Periodicity

Continuous: $\cos(\Omega t)$ is always periodic.

Discrete: $\cos(\omega n)$ is periodic **only if** f is a rational number.

Derivation 1.2.1

For periodicity with period N ($N > 0$, integer):

$$x(n + N) = x(n)$$

$$\cos(2\pi f(n + N) + \theta) = \cos(2\pi fn + \theta)$$

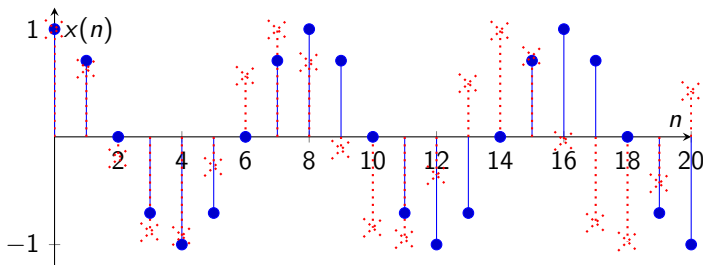
This requires $2\pi fN$ to be a multiple of 2π :

$$2\pi fN = 2\pi k \quad \Rightarrow \quad f = \frac{k}{N}$$

Conclusion: Frequency f must be a rational number.

Visualizing Discrete Periodicity

Periodic vs Aperiodic Samples



Blue: $f = 1/8$. Repeats every 8 samples.

Red (dotted): $f \approx \sqrt{2}/10$. Samples never land on the exact same value twice.

Property 1.2.5: Aliasing (Indistinguishability)*

In continuous time, distinct frequencies make distinct signals. In discrete time, frequencies separated by integer multiples of 2π are **identical**.

Derivation 1.2.2

$$\cos((\omega + 2\pi)n + \theta) = \cos(\omega n + 2\pi n + \theta)$$

Since n is an integer, $2\pi n$ is a multiple of 2π :

$$= \cos(\omega n + \theta)$$

The Implication

The sequence $x(n)$ cannot distinguish between frequency ω and $\omega + 2\pi$.

The Fundamental Range

Because of indistinguishability, we only need to analyze frequencies in a fundamental range.

Variable	Range	Alternative
ω (rad/sample)	$-\pi \leq \omega \leq \pi$	$0 \leq \omega \leq 2\pi$
f (cycles/sample)	$-1/2 \leq f \leq 1/2$	$0 \leq f \leq 1$

Frequencies outside this range are "aliases" of frequencies inside this range.

Property 1.2.6: Rate of Oscillation

Continuous: As $\Omega \rightarrow \infty$, oscillation gets faster.

Discrete:

- Lowest rate of oscillation: $\omega = 0$ (DC).
- Highest rate of oscillation: $\omega = \pi$ (or $f = 1/2$).

Why? Because as ω increases from π to 2π , it acts like a negative frequency from $-\pi$ to 0 (slowing down). $\epsilon > 0$ small.

$$\cos((\pi + \epsilon)n) = \cos((\pi - \epsilon)n \pm \text{phase})$$

Energy vs. Power Signals

Signals can be classified based on their total **Energy** (E) or average **Power** (P).

For a discrete-time signal $x(n)$:

Definition 1.2.1 (Energy)

$$E = \sum_{n=-\infty}^{\infty} |x(n)|^2$$

Definition 1.2.2 (Average Power)

$$P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x(n)|^2$$

- **Energy Signal:** $0 < E < \infty$. (Consequently, $P = 0$). Typical for transient/finite-duration signals.
- **Power Signal:** $0 < P < \infty$. (Consequently, $E = \infty$). Typical for periodic or random signals.

Comparison: Energy vs. Power Signals

Based on E and P , a signal is strictly classified into one of three categories:

1. Energy Signals

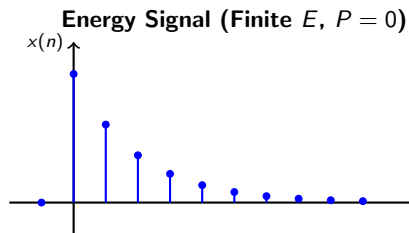
- **Condition:** $0 < E < \infty$
- **Consequence:** Average Power $P = 0$.
- **Characteristics:** Must have finite duration or asymptotically decay to zero as $n \rightarrow \pm\infty$.
- **Examples:** A single rectangular pulse, exponential decay $x(n) = (0.5)^n u(n)$.

2. Power Signals

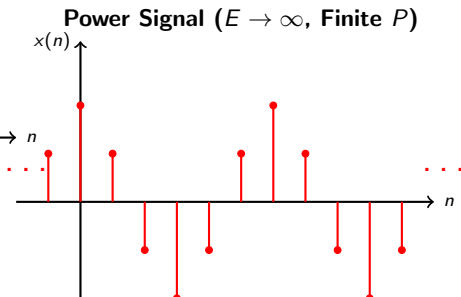
- **Condition:** $0 < P < \infty$
- **Consequence:** Total Energy $E = \infty$.
- **Characteristics:** Typically infinite in duration.
- **Examples:** Periodic sinusoids $x(n) = \cos(\omega n)$, unit step functions $u(n)$, stationary random noise.

*Note: A signal can be **neither** (e.g., $x(n) = n$, where both $E \rightarrow \infty$ and $P \rightarrow \infty$), but a signal can **never** be both an energy and a power signal simultaneously.*

Visualizing Energy vs. Power Signals



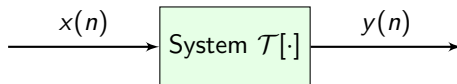
e.g., Decaying transient
 $x(n) = a^n u(n)$, $|a| < 1$



e.g., Infinite periodic sequence
 $x(n) = A \cos(\omega n)$

Systems

A discrete-time **system** is a mathematical transformation or operator that maps an input sequence $x(n)$ into an output sequence $y(n)$.



$$y(n) = \mathcal{T}[x(n)]$$

To analyze and design systems, we categorize them based on fundamental properties: **Linearity, Time-Invariance, Causality, and Stability.**

Example 1.2.1: Moving Average Filter

A 3-point moving average filter smooths data by averaging each input with the two previous inputs.

System Equation

$$y(n) = \frac{1}{3} [x(n) + x(n-1) + x(n-2)]$$

Applying the system:

- Let the input sequence $x(n)$ be a series of daily temperature readings:

$$x = \{\dots, 10, 13, \underset{n=2}{16}, \underset{n=3}{14}, 11, \dots\}$$

- To find the output at time $n = 3$, we apply the formula:

$$y(3) = \frac{1}{3} [x(3) + x(2) + x(1)]$$

$$y(3) = \frac{1}{3} [14 + 16 + 13] = \frac{43}{3} \approx 14.33$$

- The output value $y(3)$ is the smoothed temperature at that point in time.

Property 1.2.7: Linearity*

A system is **linear** if it satisfies the principle of **Superposition**, which includes both additivity and homogeneity (scaling).

If $\mathcal{T}[x_1(n)] = y_1(n)$ and $\mathcal{T}[x_2(n)] = y_2(n)$, then for any arbitrary constants a and b :

Linearity Condition

$$\mathcal{T}[ax_1(n) + bx_2(n)] = ay_1(n) + by_2(n)$$

- **Linear Example:** $y(n) = 3x(n) + x(n-1)$
- **Non-Linear Example:** $y(n) = x^2(n)$ or $y(n) = x(n) + 5$

Example 1.2.2: Linearity Test

Test the system $\mathcal{T}[x(n)] = y(n) = 3x(n)$ for linearity.

❶ **Define individual outputs:**

- For input $x_1(n)$, the output is $y_1(n) = 3x_1(n)$.
- For input $x_2(n)$, the output is $y_2(n) = 3x_2(n)$.

❷ **Calculate the Left-Hand Side (LHS):** Apply the system to a weighted sum of inputs.

$$\begin{aligned}\mathcal{T}[ax_1(n) + bx_2(n)] &= 3 \cdot (ax_1(n) + bx_2(n)) \\ &= 3ax_1(n) + 3bx_2(n)\end{aligned}$$

❸ **Calculate the Right-Hand Side (RHS):** Create a weighted sum of the individual outputs.

$$\begin{aligned}ay_1(n) + by_2(n) &= a(3x_1(n)) + b(3x_2(n)) \\ &= 3ax_1(n) + 3bx_2(n)\end{aligned}$$

Conclusion

Since LHS = RHS, the system $y(n) = 3x(n)$ is **linear**.

Property 1.2.8: Time-Invariance

A system is **Time-Invariant (TI)** if its characteristics do not change over time. A delay in the input signal causes an identical delay in the output signal.

If $y(n) = \mathcal{T}[x(n)]$, then for any time shift k :

Time-Invariance Condition

$$\mathcal{T}[x(n - k)] = y(n - k)$$

- **TI Example:** $y(n) = x(n) - x(n - 1)$
- **Time-Variant Example:** $y(n) = n \cdot x(n)$ (Scaling factor depends on current time n)

Example 1.2.3: Time-Invariance Test

Test the system $\mathcal{T}[x(n)] = y(n) = n \cdot x(n)$ for time-invariance.

① **Find the shifted output, $y(n - k)$:**

- Start with the output equation: $y(n) = n \cdot x(n)$.
- To find $y(n - k)$, replace every instance of n with $(n - k)$:

$$y(n - k) = (n - k) \cdot x(n - k)$$

② **Find the output for a shifted input, $\mathcal{T}[x(n - k)]$:**

- The system operation is to multiply the *entire* input signal by the time index n .
- Our input is now $x(n - k)$.

$$\mathcal{T}[x(n - k)] = n \cdot [x(n - k)]$$

Conclusion

We compare the two results:

$$(n - k) \cdot x(n - k) \neq n \cdot x(n - k)$$

Since $y(n - k) \neq \mathcal{T}[x(n - k)]$, the system is **time-variant**.

Property 1.2.9: Causality & Property 1.2.10: BIBO Stability

Causality: A system is **causal** if its output at time n depends *only* on present and past inputs ($x(n), x(n-1), \dots$), and not on future inputs ($x(n+1), \dots$).

- **Causal:** $y(n) = x(n) + x(n-1)$
- **Non-Causal:** $y(n) = x(n+1)$ (Requires looking into the future).

Stability (BIBO): Bounded-Input Bounded-Output stability implies that if the input is finite, the output must also be finite.

BIBO Condition

If $|x(n)| \leq M_x < \infty$ for all n , then $|y(n)| \leq M_y < \infty$ for all n .

Exercise 1.2.1: Discrete Periodicity

Problem: Determine if the following discrete-time signals are periodic. If yes, find the fundamental period N .

$$x_1(n) = \cos\left(\frac{3\pi}{4}n\right)$$

$$x_2(n) = \cos(3n)$$

Solution 1.2.1: Discrete Periodicity

Recall: $x(n)$ is periodic only if frequency f is rational ($f = k/N$).

$$\omega = 2\pi f \implies f = \frac{\omega}{2\pi}$$

1. Analysis of $x_1(n) = \cos\left(\frac{3\pi}{4}n\right)$:

$$\omega = \frac{3\pi}{4} \implies f = \frac{3\pi/4}{2\pi} = \frac{3}{8}$$

- Is f rational? **Yes.**
- Fundamental Period N : Denominator (after simplifying) is **8**.
- **Result: Periodic with $N = 8$.**

2. Analysis of $x_2(n) = \cos(3n)$:

$$\omega = 3 \implies f = \frac{3}{2\pi}$$

- Is f rational? **No** (because π is irrational).
- **Result: Aperiodic.**

Summary Part 2

- **Periodicity & Aliasing:** Discrete sinusoids require rational frequencies to be periodic. Frequencies outside $[-\pi, \pi]$ alias into this range.
- **Energy vs. Power:** Signals are categorized by finite Energy (transient) or finite Power (periodic/random).
- **System Definition:** An operator $\mathcal{T}[\cdot]$ mapping input $x(n)$ to output $y(n)$.
- **System Properties:**
 - **Linearity:** Superposition holds ($ax_1 + bx_2 \rightarrow ay_1 + by_2$).
 - **Time-Invariance:** Shifted input yields equivalently shifted output.
 - **Causality:** Output depends only on present/past inputs.
 - **Stability:** Bounded input guarantees bounded output (BIBO).

End of Lecture 1.